Databricks Data Engineering and AIOps – Course Outline

1. Duration

- 20 Hours (5 Half-Days)

2. Objectives

At the end of this workshop, participants will be able to:

- Understand how Databricks supports modern data engineering, ETL, and operational analytics workflows in a unified platform.
- Build a practical foundation in Databricks workspace usage, notebooks, compute, SQL, and Spark-based processing for ETL use cases.
- Organize telecom KPI, event, and incident data into analysis-ready layers for downstream operational use.
- Understand the role of AIOps in moving from reactive monitoring to predictive and context-aware operations.
- Explore how classical AI/ML can support anomaly detection, forecasting, and outage-oriented prediction.
- Understand how GenAI can complement AIOps through incident summarization, recommendation support, contextual interpretation, and guided operational response.
- Design a simple, repeatable AIOps workflow covering prediction, recommendation, and self-healing-oriented action patterns.

3. Audience

Data Engineers, ETL Developers, Analytics Engineers, Telecom Operations Analysts, AIOps Engineers, and technical professionals who want focused exposure to Databricks ETL and practical AIOps concepts.

4. Pre-requisite

- Basic understanding of SQL and tabular data.
- Familiarity with ETL concepts and operational data workflows.
- General awareness of KPIs, alerts, incidents, or service monitoring is helpful.
- Basic Python knowledge is useful for notebook-based exercises, but not mandatory.
- Exposure to telecom or operations data is useful, but not mandatory.

5. Hardware & Network Requirements

- Desktop/Laptop with minimum 16 GB RAM.
- Stable Internet connection.
- Browser access to the training environment.
- Local Admin Access for required setup.

6. Software Requirements

- Windows / Linux / Mac OS.
- Latest Chrome or Edge browser.
- VS Code or equivalent editor.
- Python 3.x.
- Access to Databricks workspace for labs.

7. Outline

Day 1

Module-1: Introduction to Databricks and Lakehouse Concepts

- Why organizations are adopting Databricks for data engineering, ETL, and operational analytics.
- Core platform overview: workspace, notebooks, compute, jobs, SQL, Spark, and Delta-based processing.
- Introduction to lakehouse thinking and curated data-layer design.
- Hands-on: Workspace walkthrough, notebook execution, sample data exploration, and a simple ETL-oriented transformation.

Day 2

Module-2: Data Engineering Foundations on Databricks

- Ingestion, transformation, curation, and optimization fundamentals.
- Structuring raw, cleaned, and business-ready datasets for telecom operations data.
- Preparing KPI, event, and incident-related datasets for downstream analysis.
- Hands-on: Load sample telecom-aligned data, create curated layers, and prepare analysis-ready tables.

Day 3

Module-3: Telecom Data Exploration for AIOps

- Understanding telecom KPI categories such as latency, utilization, failure indicators, quality signals, and customer-impact metrics.
- Bringing together KPI data, alerts, events, and incidents for operational analysis.
- Using notebooks and SQL for exploratory analysis, trend review, rolling windows, and correlation checks.
- Hands-on: Explore degradation and outage-related patterns to identify useful signals for AIOps.

Day 4

Module-4: AIOps with Classical AI/ML

- What AIOps means in practical terms and how it improves on purely reactive monitoring.
- Anomaly detection, forecasting, and outage-oriented prediction concepts.
- Framing prediction problems using features, labels, windows, and evaluation logic.
- Hands-on: Build a simple prediction-oriented workflow using curated KPI and event data.

Day 5

Module-5: Recommendation, GenAI, and Self-Healing Workflow

- Turning anomaly and prediction outputs into recommendation-oriented operational actions.
- Using GenAI for incident summarization, contextual interpretation, action guidance, and analyst assistance.
- Understanding self-healing concepts such as alert enrichment, approval-based actions, and controlled remediation workflows.
- Hands-on: Guided end-to-end mini workshop covering ETL preparation, prediction review, recommendation flow, and self-healing discussion.